SUEZ Platform – Product Requirements Document (PRD)

Version: 1.0

Status: Draft

Owner: Product / Platform Architecture

Audience: Engineering, Security, SRE, Platform Ops, Leadership

1. Product Overview

SUEZ is an Enterprise SaaS platform for governed AI usage. It provides centralized control, policy enforcement, security, cost visibility, and auditing for enterprise AI consumption across multiple providers and tools.

The platform acts as a secure gateway and control plane between enterprise applications and AI providers.

Core Goals

• Secure enterprise AI usage

• Enforce governance policies

• Provide cost visibility and billing

• Enable multi-tenant SaaS deployment

• Support enterprise authentication and compliance

2. Problem Statement

Enterprises adopting AI face several challenges including uncontrolled AI access, lack of governance, shadow AI usage, limited auditing, and unpredictable AI costs.

SUEZ solves these problems by providing a centralized governance platform for AI access.

3. Product Vision

SUEZ becomes the enterprise control plane for AI usage providing governance, cost control, security enforcement, provider abstraction, and auditability.

4. Key Personas

Platform Administrator

Manages tenants, providers, billing configuration, and overall system operations.

Tenant Administrator

Manages tenant users, policies, API keys, and usage monitoring.

Application Developer

Integrates applications with the AI gateway and uses provisioned models, tools, and agents.

Security / Compliance Officer

Ensures AI usage follows compliance, governance policies, and security controls.

5. Platform Capabilities

The platform provisions and governs AI resources including Models, Tools, and AI Agents.

Resource Hierarchy

Tenant → Models → Tools → Agents (Agents use models and tools governed by policy).

6. Model Provisioning

Administrators can provision AI models available to tenants.

Supported providers may include OpenAI, Anthropic, AWS Bedrock, Google Vertex AI, Azure OpenAI, and local models.

Model Configuration Fields

model_id, provider, model_name, tenant_scope, token_limit, cost_profile, enabled

Governance

Policies can enforce allowed models, token limits, provider restrictions, and prompt size limits.

7. Tool Provisioning

Tools represent external capabilities that AI models or agents may invoke.

Examples include web search, knowledge base retrieval, database queries, and internal APIs.

Tool Configuration Fields

tool_id, name, endpoint, auth_type, permissions, tenant_scope

Security Requirements

Tools must support scoped credentials, rate limits, policy enforcement, and audit logging.

8. AI Agent Provisioning

Agents are orchestrated AI workflows composed of models, tools, policies, and optional context or memory.

Agent Configuration

agent_id, tenant_id, model, tools, policies, system_prompt, max_steps

Agent Execution Flow

Application → Gateway → Agent Runtime → Model + Tools

9. Usage Metering

Each request generates a usage record including tenant_id, user_id, resource_type, resource_id, tokens, cost, and timestamp.

Agent usage also records steps executed, tools invoked, and model calls.

10. APIs

Model APIs

POST /models | GET /models | DELETE /models/{id}

Tool APIs

POST /tools | GET /tools | DELETE /tools/{id}

Agent APIs

POST /agents | GET /agents | DELETE /agents/{id}

11. Non■Functional Requirements

Security

TLS everywhere, secure API key storage, IAM isolation, tenant isolation, and compliance readiness.

Scalability

Initial target: 100 tenants. Scaling target: 10,000 tenants.

Traffic target: 1000 requests/sec with horizontal scaling.

Reliability

Availability target: 99.9%. RTO: 1 hour. RPO: 15 minutes.

Performance

Gateway latency overhead target <150ms. Policy evaluation <50ms.

12. Observability

Metrics: request rate, latency, error rate, quota usage.

Logs: request logs, policy decisions, admin actions.

Tracing: request pipeline across services.

13. Billing

Billing supports subscription plans and usage-based billing.

Billing units: model tokens, tool invocations, agent execution steps.

14. Future Roadmap

Planned enhancements include agent marketplace, tool registry, model benchmarking, AI risk scoring, and multi-region deployment.